



Mustang Math Tournament 2025



Focused Filly Foal Round



Basic Format

- This round contains 20 problems to be solved in 60 minutes.
- The problems are divided into four sets (Algebra, Geometry, Number Theory, Combinatorics) with five problems each.
- The more problems you solve in a single set, the more points you will receive per problem. You receive 4 points for the first problem solved, 5 points for the next problem you solve, and so on up to 8 points if you solve the last problem.
- For instance, if you solve two problems of the Algebra set, you will receive 9 points total for that set (regardless of which two you solved). This makes for a maximum of 30 points per set.
- **Do not** write below the provided answer blank inside each space on the answer sheet (the space is for grading purposes).



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**WE WILL NOT GRADE ANY WORK ON THIS
PAGE.**



Algebra (A)

- [A1]. Jeremy takes two different digits from the number 173205 and multiplies them. What is the largest possible result?
- [A2]. The set of positive integers $\{4, 4, a, b, c\}$ has no unique mode, and the mean and median are equal. What is the least possible value of $a + b + c$?
- [A3]. In a big room of animals, there are pigs, which have 4 legs; cows, which have 4 legs; and chickens, which have 2 legs. When the maximum number of chicken-cow pairs are formed, they have 66 legs total, and when the maximum number of chicken-pig pairs are formed, they have 42 legs total. If there are 16 cows in the room, how many animals are in the room?
- [A4]. Find the number of ordered pairs of integers (x, y) , each with absolute value less than or equal to 20 such that

$$\frac{|x| + |y|}{||y| - x|} \cdot \frac{x + y}{||x| - y|} = -1$$

- [A5]. A function $f(x)$ defined over the integers satisfies the following property:

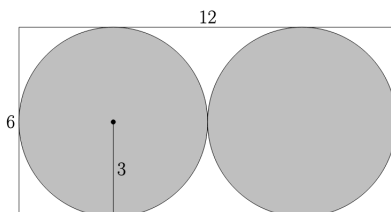
$$f(x + y) = 17^x f(y) + 17^y f(x)$$

Given that $f(2) = 2023$, find the last two digits of $f(6)$.

Geometry (G)

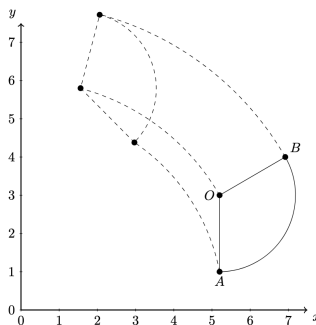
[G1]. Square $ABCD$ has a point E on CD such that $AE = 13$ and $DE = 5$. What is BE^2 ?

[G2]. John has a CD box that is 6 inches by 12 inches. He has two CDs of radius 3 inches, placed flat on the bottom without overlap. What is the radius, in inches, of the largest CD he can fit in the box without overlapping either of the first two CDs? Express your answer as a common fraction.



[G3]. In triangle ABC , point M is the midpoint of BC , and point P is on segment $\overline{AM}$ such that $AP = PM = CP = BM = 2$. What is AB ? Express your answer in simplest radical form.

[G4]. Points $O(3\sqrt{3}, 3)$, $A(3\sqrt{3}, 1)$, and $B(4\sqrt{3}, 4)$ form a circular sector centered at point O on the coordinate plane. This sector is rotated 45° counter-clockwise about the origin. What is the area of the path traced out by the sector, in terms of π ?



[G5]. Let parallelogram $ABCD$ have $AB = 13$, $BC = 5$, and $AC < BD$. Let line $\overline{AD}$ and side $\overline{CD}$ intersect the circumcircle of $\triangle ABC$ at points $P \neq A$ and $Q \neq C$, respectively. Given that $\overline{PQ} \perp \overline{AB}$, find PQ . Express your answer as a common fraction.



Number Theory (N)

- [N1]. Let a and b be prime numbers such that $a + b = 50$. Find the greatest possible value of ab .
- [N2]. What is the largest power of 2 that can divide a 4-digit number of the form $\overline{ABAB}$ for nonzero digits A and B ?
- [N3]. Alice and Bob are thinking of the numbers 2025 and 5022, respectively. They both write down the remainder when their number is divided by the same positive integer s . Given that the numbers they write down are equivalent and add up to a multiple of s , what is the maximum value of s ?
- [N4]. Let S be the set $\{3, 4, 5, 6, 7, 8\}$. Albert selects a non-empty subset of these 6 integers and multiplies his numbers together. Using this product, Fiona tries to determine which of the 6 integers were selected. What is the sum of all possible products for which it is impossible for Fiona to determine the integers selected?
- [N5]. Let a, b , and c be positive integers satisfying $\gcd(a, b) = 1296$, $\gcd(b, c) = 15$, and

$$\frac{\gcd(a, c)}{1296} > \left(\frac{\gcd(a, b)}{\gcd(a, c)}\right)^2.$$

How many possible values of c are there for $c < 1000$?



Combinatorics (C)

- [C1]. Albert, Betty, and Carlos are 3 students in a class of 24 total students. The teacher randomly divides the class into 4 teams with 6 students in each. Given that Albert is not in the same team as Betty and Betty is not in the same team as Carlos, what is the probability that Carlos is in the same team as Albert? Express your answer as a common fraction.
- [C2]. How many distinct words can be made by a rearrangement of the word “MUSTMATH”, including the word “MUSTMATH” itself?
- [C3]. Aiden, Brayden, Caden, and Dayden are playing a bracket-style tournament for a 2-person game. The 1st round pairings are Aiden vs. Brayden and Caden vs. Dayden, and the winners of each game will play each other in the 2nd round to determine the tournament champion. There are no draws, and every player has a $\frac{1}{2}$ chance of winning their game, except if the matchup is Brayden vs. Caden, in which case Brayden will win with probability $\frac{4}{5}$. What is the probability that somebody other than Brayden wins the tournament? Express your answer as a common fraction.
- [C4]. There are 12 indistinguishable apples to be split amongst 5 children. To make sure there are no fights, no child can get more apples than the other four combined. How many ways are there to distribute the apples without fights breaking out?
- [C5]. What is the expected value of $1@2@3@4@5$, where each @ is either multiplication or division, with each operation being equally likely? Express your answer as a common fraction.



Acceptable Answers

The following rules provide guidelines for acceptable answers in this round. Please note that any specifications provided in a problem will take precedence over these rules. The decisions of MMT coordinators are final.

- Common fractions are defined as a fraction in the form $\pm \frac{a}{b}$ where a and b are natural numbers and $\gcd(a, b) = 1$.
- Ratios and fractional answers should be expressed as common fractions unless otherwise specified.
- Radicals should be simplified. A simplified radical must satisfy:
 - No square factors, fractions, or nested radicals inside a radical
 - No radicals inside the denominator of a fraction
- Answers must be expressed to the exact accuracy called for in the problem (e.g. 25.0 will not be accepted for 25 and 25 will not be accepted for 25.0).
- Do not make approximations for numbers (e.g. 3.14 or $\frac{22}{7}$ for π) unless otherwise specified.
- Units **do not** need to be included but must be correct if included.